

Text Summarization

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Text Summarization

- Subfield of Natural Language Processing (NLP) and Computational Linguistics
- Primary goal is to automatically produce a concise, coherent, and fluent summary of a longer text document, capturing the most critical information and main ideas.

Types Based on the Nature of the Summary (Output)

- Extractive Summarization
- Abstractive Summarization

Extractive Summarization

- This method works by identifying and extracting the most "important" sentences, phrases, or sections directly from the source text and concatenating them to form a summary. It does not generate any new text.
- It's like using a highlighter to mark the key sentences in a document and then reading only those highlighted parts.

Extractive Summarization Algorithm

- Scores sentences based on their importance. Importance can be determined by factors like:
- Frequency of words (e.g., words that appear often are likely important).
- Position of the sentence (e.g., the first and last sentences of a paragraph are often topic and concluding sentences).
- Presence of title words (sentences containing words from the title or headings).
- Linguistic features (e.g., sentences with cue phrases like "in conclusion")

Extractive Summarization Algorithm

- Ranks the sentences according to their scores.
- Selects the top-N highest-ranking sentences to form the summary.
- **Pros:** Simpler, more straightforward, and the facts are guaranteed to be accurate to the source text.
- **Cons:** The summary can be redundant, lack flow between sentences, and miss nuanced connections that require paraphrasing.

Abstractive Summarization

- This is a more advanced technique that aims to understand the core meaning of the text and then express that meaning in new, concise language, much like a human would. It involves paraphrasing and generating novel sentences.
- It's like reading a document, understanding it, and then writing a summary in your own words.

Abstractive Summarization

- Abstractive summarization relies on sophisticated deep learning models, particularly Sequence-to-Sequence (Seq2Seq) models with attention mechanisms (like Transformers, which power models like BART, T5, and GPT). These models:
- Encode the source text into an internal representation that captures its semantic meaning.
- Decode this representation to generate a new, shorter sequence of words that constitutes the summary.

Abstractive Summarization

- Pros: Can produce more fluent, concise, and coherent summaries that read naturally.
- Cons: Computationally much more complex, requires massive amounts of training data, and runs the risk of introducing factual inaccuracies or "hallucinating" content not present in the source.

Frequency-Based Extractive Summarization

- Importance of a word is determined by how often it appears in the text
- Sentences containing a high concentration of these frequent words are considered important and are extracted for the summary.

How it works

- Preprocessing: The text is cleaned and normalized.
- Tokenization: Split the text into individual words (tokens).
- Stopword Removal: Remove very common words that carry little meaning (e.g., "the," "a," "is," "in," "and").
- Stemming/Lemmatization: Reduce words to their base or root form (e.g., "running," "ran" -> "run").

How it works

- **Frequency Calculation:** Calculate the frequency of each remaining word in the preprocessed text.
- **Sentence Scoring:** Each sentence in the original text is given a score.
- The score is typically the sum of the frequencies of the important words (non-stopwords) it contains.
- Sometimes, it's the average frequency of the content words.

How it works

- **Summary Generation:** Select the top N sentences with the highest scores to form the summary. The order is usually based on their original appearance in the text to maintain coherence.

Example

- **Original Text:**
- "Cats are popular pets. Many people own cats. Cats are independent animals. Dogs are also popular pets. Dogs are known for their loyalty."

Preprocessing & Frequency

- After stopword removal: [cats, popular, pets, many, people, own, cats, cats, independent, animals, dogs, popular, pets, dogs, known, loyalty]

Word Frequencies:

- cats: 3, popular: 2, pets: 2, dogs: 2
- many: 1, people: 1, own: 1, independent: 1, animals: 1, known: 1, loyalty: 1

Sentence Scoring

- "Cats are popular pets." -> Score: $\text{cats}(3) + \text{popular}(2) + \text{pets}(2) = 7$
- "Many people own cats." -> Score: $\text{many}(1) + \text{people}(1) + \text{own}(1) + \text{cats}(3) = 6$
- "Cats are independent animals." -> Score: $\text{cats}(3) + \text{independent}(1) + \text{animals}(1) = 5$
- "Dogs are also popular pets." -> Score: $\text{dogs}(2) + \text{popular}(2) + \text{pets}(2) = 6$
- "Dogs are known for their loyalty." -> Score: $\text{dogs}(2) + \text{known}(1) +$

Summary (Top 2 Sentences):

- "Cats are popular pets. Dogs are also popular pets."

Pros and Cons:

- Pros: Very simple to implement and fast to compute.
- Cons: Can be simplistic. It may miss semantically important sentences that don't contain the most frequent words. It can also be biased towards longer sentences.

Query (Q) Phrase-Based Extractive Summarization

- This method is goal-oriented. Instead of finding a general summary, it creates a summary specific to a user's query or area of interest (the "q phrase").
- The importance of a sentence is determined by its relevance to the query.

How it works

- **Define the Query:** The user provides a query phrase (e.g., "animal loyalty").
- **Preprocessing:** Both the query and the source text are preprocessed (tokenization, stopword removal, stemming/lemmatization).
- **Similarity Scoring:** Each sentence in the text is scored based on its similarity to the query.
- A simple and common method is to use the TF-IDF representation of the sentence and the query, and then calculate the cosine

How it works

- **TF-IDF (Term Frequency-Inverse Document Frequency)** gives higher weight to words that are frequent in the current sentence but rare in the entire document (or a large corpus), making them good discriminators.
- **Cosine Similarity** measures the angle between the TF-IDF vectors of the sentence and the query. A higher cosine score means the sentence is more relevant to the query.
- **Summary Generation:** Select the top N sentences with the highest similarity scores to the query.

Example using the same text:

- **User Query:** "animal loyalty"
- **Preprocessing:** The query becomes the vector [animal, loyalty].

Sentence Scoring (Conceptual TF-IDF/Cosine Similarity)

- We look for sentences that contain the words "animal" and "loyalty."
- **Sentence 3:** "Cats are independent animals." -> Contains "animal." Moderate score.
- **Sentence 5:** "Dogs are known for their loyalty." -> Contains "loyalty." Moderate score.
- **Sentence 1, 2, 4:** Do not contain the query words. Very low score.

Query-Based Summary (Top 2 Sentences):

- "Cats are independent animals. Dogs are known for their loyalty."
- Notice how this summary is completely different from the frequency-based one because it answers a specific question.

Pros and Cons:

- **Pros:** Highly relevant to the user's specific information need.
Produces focused, query-driven summaries.
- **Cons:** Requires a well-defined query. The quality of the summary is entirely dependent on the quality of the query.

Metric for text summarization

- Reference-based (comparing to a human-written summary) and
- Reference-free (evaluating the summary on its own merits).

Metric for text summarization

Metric	Type	Key Idea	Pros	Cons
ROUGE	Reference-based	Lexical Overlap	Fast, standard, easy to use	Poor at capturing meaning, recall-biased
BERTScore	Reference-based	Semantic Similarity	Robust to paraphrasing, better correlation	Slower than ROUGE
BLEURT	Reference-based	Learned Metric	State-of-the-art correlation with humans	Requires training data
Factuality Metrics	Reference-free	Checks against source	Evaluates a critical quality	Often complex to set up
LLM-as-a-Judge	Reference-free	Uses powerful LLM	Highly flexible, evaluates multiple dimensions	Expensive, black-box, slow
Human Evaluation	N/A	Human Judgment	Gold standard, reliable	Very expensive, slow, not scalable

Reference-Based Metrics (Automatic)

- These are the most common metrics, as they are cheap and fast to compute automatically.
- **A. Lexical Overlap Metrics (N-gram based)**
- These measure the word-for-word or phrase-for-phrase overlap between the generated summary and a reference (human-written) summary.

Reference-Based Metrics (Automatic)

- ROUGE (Recall-Oriented Understudy for Gisting Evaluation): The industry standard. It works by counting the overlapping units like n-grams, word sequences, etc.
- **ROUGE-N**: Overlap of N-grams.
- **ROUGE-1**: Overlap of unigrams (single words).
- **ROUGE-2**: Overlap of bigrams (two-word sequences). This is very common.
- **ROUGE-L**: Based on the longest common subsequence (LCS). It captures sentence-level structure and is less sensitive to word

Reference-Based Metrics (Automatic)

- **Pros:** Fast, automatic, widely used, correlates reasonably well with human judgment for extractive summaries.
- **Cons:** Poor at evaluating meaning and fluency. A summary can have high ROUGE scores by just copying phrases without making sense. It's recall-oriented, which can penalize concise summaries.

Reference-Based Metrics (Automatic)

- **BLEU (Bilingual Evaluation Understudy):**
- Originally designed for machine translation, it measures precision (how many words in the generated summary appear in the reference). It's less common for summarization than ROUGE.
- **Pros:** Precision-focused.
- **Cons:** Not ideal for summarization where recall (capturing key points) is often more important.

BLEU vs ROUGE

- **BLEU:** How many generated words are correct?
- **ROUGE:** How much of the reference did you capture?

Numerical Problem on BLEU-1 and ROUGE-1

Reference Sentence (Ground Truth)

- R: "the cat is on the mat"

Candidate Sentence (Generated Output)

- C: "the cat sat on the mat"

Total words

- Reference length = 6
- Candidate length = 6

Calculate BLEU-1 (Unigram Precision)

- BLEU measures precision — how many words in the candidate appear in the reference.

Candidate unigrams

- the, cat, sat, on, the, mat

Reference unigrams

- the, cat, is, on, the, mat

Calculate BLEU-1 (Unigram Precision)

- Count overlapping words

Word	Candidate Count	Reference Count	Clipped Count
the	2	2	2
cat	1	1	1
sat	1	0	0
on	1	1	1
mat	1	1	1

Total clipped matches = 5

BLEU-1 Precision - $5/6 = 0.83$

Calculate ROUGE-1 (Recall)

- ROUGE measures recall — how much of the reference is captured by the candidate.
- **Overlapping words**
- Same overlap as before = 5
- Reference total words - 6
- ROUGE-1 Recall - $5/6 = 0.83$

Change the Candidate Sentence

BLEU-1

Overlap words = the, cat, on, mat = 4

$$\text{BLEU} = 4/4 = 1.0$$

ROUGE-1

$$\text{ROUGE} = 4/6 = 0.67$$

Change the Candidate Sentence

Candidate 2

- "the cat on mat"
- Candidate length = 4

Change the Candidate Sentence

BLEU-1

Overlap words = the, cat, on, mat = 4

$$\text{BLEU} = 4/4 = 1.0$$

ROUGE-1

$$\text{ROUGE} = 4/6 = 0.67$$

BLEU-2 and ROUGE-L

Reference Sentence (R)

the quick brown fox jumps over the lazy dog

Candidate Sentence (C)

the quick fox jumps over lazy dog

BLEU-2 and ROUGE-L

Sentence Lengths

Reference length

$$|R| = 9$$

Candidate length

$$|C| = 7$$

BLEU-2 and ROUGE-L

Unigram Precision

Candidate unigrams: the, quick, fox, jumps, over, lazy, dog

Reference unigrams: the, quick, brown, fox, jumps, over, the, lazy, dog

Matching words: the, quick, fox, jumps, over, lazy, dog

Total matches = 7

$P1 = 7/7 = 1.0$

BLEU-2 and ROUGE-L

Bigram Precision

Candidate Bigrams

the quick

quick fox

fox jumps

jumps over

over lazy

lazy dog

Total = 6

BLEU-2 and ROUGE-L

Bigram Precision

Reference Bigrams

the quick

quick brown

brown fox

fox jumps

jumps over

over the

the lazy

lazy dog

BLEU-2 and ROUGE-L

Bigram Precision

Matching Bigrams

the quick

fox jumps

jumps over

lazy dog

Matches = 4

$P_2 = 4/6 = 0.667$

BLEU-2 and ROUGE-L

BLEU-2 Score

BLEU-2 uses geometric mean:

$$BLEU = \sqrt{P_1 \times P_2}$$

$$BLEU = \sqrt{1.0 \times 0.667}$$

$$BLEU = \sqrt{0.667} = 0.816$$

BLEU-2 $\approx$ 0.82

BLEU-2 and ROUGE-L

ROUGE-L (Longest Common Subsequence)

Reference: the quick brown fox jumps over the lazy dog

Candidate: the quick fox jumps over lazy dog

Longest Common Subsequence: the quick fox jumps over lazy dog

Length of LCS: $LCS = 7$

BLEU-2 and ROUGE-L

Recall

$$R = \frac{LCS}{|Reference|}$$

$$R = \frac{7}{9} = 0.78$$

BLEU-2 and ROUGE-L

Precision

$$P = \frac{LCS}{|Candidate|}$$

$$P = \frac{7}{7} = 1.0$$

BLEU-2 and ROUGE-L

ROUGE-L (F1 Score)

$$F1 = \frac{2PR}{P + R}$$
$$F1 = \frac{2(1.0)(0.78)}{1.78}$$
$$F1 = 0.876$$

ROUGE-L $\approx$ 0.88

When "brown" and "the" are missing:

BLEU-2 drops because bigram matches decrease

ROUGE-L remains high because the main sentence structure is preserved

Semantic / Embedding-Based Metrics

- These are more advanced and measure semantic similarity rather than just lexical overlap.
- BERTScore:
- Uses pre-trained BERT models to compute contextual embeddings for words in both the candidate and reference summaries. It then matches words based on cosine similarity in this embedding space.
- Pros: Captures semantic similarity, more robust to paraphrasing, correlates better with human judgment than ROUGE.
- Cons: Computationally more expensive than ROUGE.

Semantic / Embedding-Based Metrics

- MoverScore:
- An extension of BERTScore that uses Word Mover's Distance, considering the entire "earth" of text and the cost of moving semantic meaning from one summary to the other.
- Pros: Can capture document-level semantics more effectively.
- Cons: Even more computationally expensive.

Semantic / Embedding-Based Metrics

- BLEURT (Bilingual Evaluation Understudy with Representations from Transformers):
- A learned metric that is fine-tuned on human ratings to predict the quality score directly. It combines the power of BERT-like models with direct supervision from human judgments.
- Pros: Currently one of the best correlates with human judgment.
- Cons: Requires training data and is model-dependent.

Reference-Free Metrics

- Crucial for real-world scenarios where a "perfect" human reference summary is not available.
- Quality Dimensions: - You can use a model to evaluate specific qualities of the summary against the source document.
- Factual Consistency: Does the summary contain information that is factually supported by the source document? (Metrics like QAFactEval)
- Relevance/Informativeness: Does the summary capture the key points of the source?
- Coherence & Fluency: Is the summary well-written, grammatically

Reference-Free Metrics

- **LLM-as-a-Judge:**
- Rapidly growing and highly effective approach. You prompt a powerful Large Language Model (like GPT-4, Claude 3) to act as an evaluator.
- **Prompt Example:** "Given the following source document and a summary, rate the summary from 1-5 on factual consistency, relevance, and coherence. Provide a justification."
- **Pros:** Extremely flexible, can evaluate nuanced aspects like conciseness and style, requires no training.
- **Cons:** Can be expensive, slow, and the evaluation itself is a black

Human Evaluation (The Gold Standard)

- Despite all automatic metrics, human evaluation remains the most reliable method. It typically assesses:
- **Coherence:** Is the summary easy to understand and logically structured?
- **Consistency:** Are the facts in the summary faithful to the source document? (This is critical and often where automatic metrics fail).
- **Fluency:** Is the summary well-written and grammatically correct?
- **Relevance:** Does the summary capture the most important information from the source?

Practical Recommendations

- For Quick & Standardized Benchmarking: Start with ROUGE (especially ROUGE-2 and ROUGE-L). It's the common language of academic papers.
- For a More Nuanced Automatic Evaluation: Use BERTScore alongside ROUGE to get a sense of both lexical and semantic similarity.
- For High-Stakes or Research Evaluation: Use LLM-as-a-Judge (e.g., with GPT-4) with a carefully designed prompt to evaluate

Practical Recommendations

- For Final Validation: Always run a limited human evaluation to spot-check the results of your automatic metrics, especially focusing on factual consistency, as this is the most common failure mode for modern summarization systems.